

Determination of volatile nitrosamines in meat products by microwave-assisted extraction and dispersive liquid-liquid microextraction coupled to ga...

Microwave-assisted extraction (MAE) and dispersive liquid-liquid microextraction (DLLME) coupled with gas chromatography-mass spectrometry (GC-MS) were evaluated for use in the extraction and preconcentration of volatile nitrosamines in meat products. Parameters affecting MAE, such as the extraction solvent used, and DLLME, including the nature and volume of the extracting and disperser solvents, extraction time, salt addition and centrifugation time, were optimized. In the MAE method, 0.25g of sample mass was extracted in 10mL NaOH (0.05M) in a closed-vessel system. For DLLME, 1.5mL of methanol (disperser solvent) containing 20 μ L of carbon tetrachloride (extraction solvent) was rapidly injected by syringe into 5mL of the sample extract solution (previously adjusted to pH 6), thereby forming a cloudy solution. Phase separation was performed by centrifugation, and a volume of 3 μ L of the sedimented phase was analyzed by GC-MS. The enrichment factors provided by DLLME varied from 220 to 342 for N-nitrosodiethylamine and N-nitrosopiperidine, respectively. The matrix effect was evaluated for different samples, and it was concluded that sample quantification can be carried out by aqueous calibration. Under the optimized conditions, detection limits ranged from 0.003 to 0.014ngmL⁽⁻¹⁾ for NPIP and NMEA, respectively (0.12-0.56ngg⁽⁻¹⁾ in the meat products). Copyright © 2011 Elsevier B.V. All rights reserved.